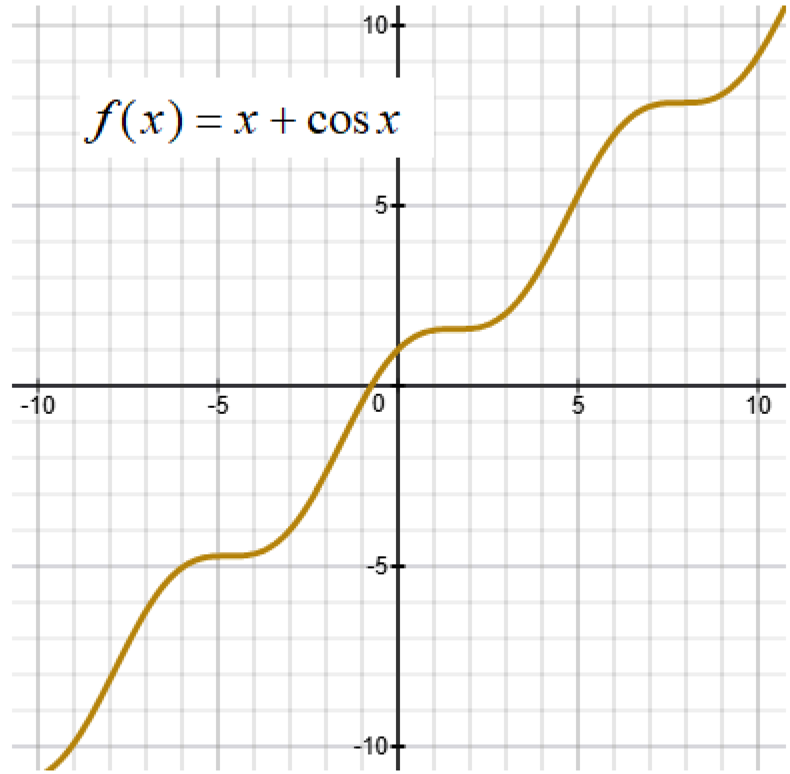
1) $f(x) = x + \cos x$

Find the x-intercept between $x = -1$ and $x = 0$.

Use initial guess of -1.



<u>Iteration 1:</u>

Set $x_1 = -1$ (initial guess)

Calculate: $t_1 = x_1 - \dfrac{f(x_1)}{f'(x_1)} = \underline{\quad ? \quad}$

Calculate: $d_1 = |x_1 - t_1| = \underline{\quad ? \quad}$

<u>Iteration 2:</u>

Set $x_2 = t_1 = \underline{\quad ? \quad}$

Calculate: $t_2 = x_2 - \dfrac{f(x_2)}{f'(x_2)} = \underline{\quad ? \quad}$

Calculate: $d_2 = |x_2 - t_2| = \underline{\quad ? \quad}$

Iteration 3:

Set $x_3 = t_2 =$ ___?___

Calculate: $t_3 = x_3 - \dfrac{f(x_3)}{f'(x_3)} =$ ___?___

Calculate: $d_3 = |x_3 - t_3| =$ ___?___

Iteration 4:

Set $x_4 = t_3 =$ ___?___

Calculate: $t_4 = x_4 - \dfrac{f(x_4)}{f'(x_4)} =$ ___?___

Calculate: $d_4 = |x_4 - t_4| =$ ___?___

The x-intercept between $x = -1$ and $x = 0$ is ___?___.